MATH 1352-008 Exam I February 8, 2008

Form B

Answer the problems on separate paper. You do <u>not</u> need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **<u>Show all relevant supporting steps!</u>**

1. (12 pts) Sketch the region bounded between the curves $y = x^2 - 5x - 3$ and $y = -2x + 7$. Find the area of that region.

2. Let R be the region, in the first quadrant, bounded by the curves $y = 3x^2$, $y = -x + 14$ and by the *y*-axis. Set up, but do **not** evaluate, an integral to compute the volume of the solid of revolution generated by revolving the region R about the indicated axis of rotation:

   a. (10 pts) the *y*-axis

   b. (10 pts) the line $y = -3$

3. (12 pts) The polar curves $r^2 = 4\sin 2q$ and $r = \sqrt{2}$ have four distinct intersection points. Find polar coordinates for two of the intersections points.

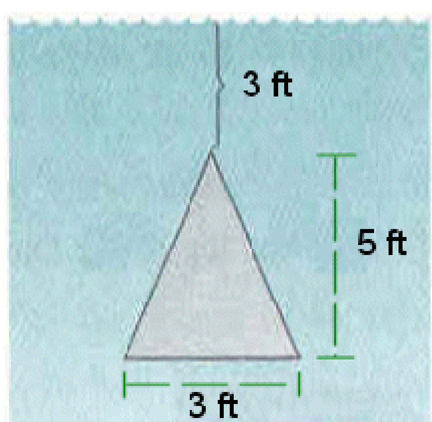
4. (12 pts) Consider the four-petal rose given by $r = 5\sin 2q$. Find the area in one of the petals. It may help to know the following:

$$\int \cos^2 u\, du = \tfrac{1}{2}u + \tfrac{1}{4}\sin 2u + c \qquad \int \sin^2 u\, du = \tfrac{1}{2}u - \tfrac{1}{4}\sin 2u + c$$

5. (12 pts) Setup, but do **not** evaluate, an integral to find the length of the curve $y = \sqrt{x+1}$ from $x = 0$ to $x = 3$.

6. (12 pts) Setup, but do **not** evaluate, an integral to find the surface area generated by revolving the curve $y = x/2 + 1/x$ from $x = 1$ to $x = 4$ about the *x*-axis.

7. (12 pts) Consider a vertical plate in tank filled with water (density $r = 62.4$) – see figure to the right. Calculate the fluid force against the face of the vertical plate.



8. (12 pts) Find the *y*-coordinate of the centroid of the planar region in the first quadrant bounded by the curves $y = 1 + 3x$ and $y = -3x + 7$ and the *y*-axis.